

MD KAMRUZZAMAN

Python Backend Developer | FastAPI | PostgreSQL | Docker | CI/CD

Dhaka, Bangladesh (Open to Remote)

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Python backend developer with a B.Tech in Computer Science and Engineering with a Machine Learning specialization. Built and deployed two practical backend and ML systems using FastAPI, PostgreSQL, Docker, CI/CD, MLflow, and Kubernetes. Experienced in REST API development, authentication, database design, automated testing, deployment, monitoring, and ML model serving. Focused on clean architecture, reliability, testing, and production-oriented backend development.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Backend: FastAPI, REST APIs, Pydantic, SQLAlchemy, Alembic

Databases: PostgreSQL, MySQL

Authentication & Security: JWT, bcrypt, OAuth2

DevOps: Docker, Docker Compose, GitHub Actions, Git, Linux

Cloud: Render, Neon

Testing: pytest, FastAPI TestClient, Ruff

ML / MLOps: scikit-learn, MLflow, NumPy, Pandas, Prometheus, Grafana, Evidently

PROJECTS

Expense Tracker API

[Code](#) | [Live Demo](#)

- Built and deployed a REST API for personal expense tracking using FastAPI with JWT authentication, bcrypt password hashing, and user-scoped data isolation enforced at the query layer.
- Designed a PostgreSQL database schema using SQLAlchemy ORM and managed database migrations with Alembic.
- Implemented pagination, category and amount filtering, Pydantic validation, structured error handling, and proper HTTP status codes.
- Achieved 91% test coverage using pytest across authentication, CRUD operations, authorization, and edge cases.
- Containerized the application with Docker and configured GitHub Actions for automated testing and Ruff linting.
- Deployed the API on Render with PostgreSQL hosted on Neon and provided a public Swagger/OpenAPI interface.

ML Expense Platform

[Code](#) | [Live Demo](#)

- Built an end-to-end ML serving platform that predicts expense categories from free-text descriptions.
- Developed a scikit-learn TF-IDF and Logistic Regression pipeline and tracked four MLflow experiments with parameter sweeps for regularization strength.
- Built a FastAPI `/predict` endpoint with Pydantic validation and lazy model loading for inference.
- Containerized the application with Docker and deployed a Kubernetes configuration with two replicas, liveness and readiness probes, and self-healing behavior.
- Added Prometheus metrics for prediction counts by category and prediction latency, with Grafana dashboards for monitoring.
- Implemented Evidently-based drift detection and verified 50% column drift using simulated production data.

EDUCATION

Lovely Professional University, Punjab, India

B.Tech in Computer Science and Engineering, Machine Learning Specialization

2025

Shahid Bulbul Government College, Bangladesh

Higher Secondary Certificate, 2018

Imam Hossain Academy, Bangladesh

Secondary School Certificate, 2016

CERTIFICATIONS

- Meta Back-End Developer** — Coursera [\[Verify\]](#)
- Natural Language Processing Specialization** — DeepLearning.AI, Coursera [\[Verify\]](#)
- Python Data Structures** — University of Michigan, Coursera [\[Verify\]](#)
- Java Programming Fundamentals** — Coursera [\[Verify\]](#)